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OBJECT EXPLORATION AND A PROBLEM WITH REDUCTIONISM

ABSTRACT. The purpose of this paper is to use neuroscientific evidence to address the philosophical issue of intertheoretic reduction. In particular, we present a literature review and a new experiment to show that the reduction of cognitive psychology to neuroscience is implausible. To make this case, we look at research using object exploration, an important experimental paradigm in neuroscience, behavioral genetics and psychopharmacology. We show that a good deal of object exploration research is potentially confounded precisely because it assumes that psychological generalizations can be reduced to neuroscientific ones.

1. INTRODUCTION

From the work of philosophers interested in the sciences of the mind, you'd get the impression that the only type of neuroscience is cognitive neuroscience. It's not hard to imagine why this is true. Cognitive neuroscientists focus on how processes in the brain might represent things outside the brain (in the body, in the environment). This has obvious connections to things philosophers are interested in: epistemology, artificial intelligence, cognitive science, and so on. But, of course, cognitive neuroscience is actually a small sliver of the research done by neuroscientists, whose focus has become increasingly molecular and genetic in recent years. John Bickle's recent work (2003) is one of the few pieces of philosophy that acknowledges this trend in the neurosciences. (Paul Thagard 2002 is another.) We applaud Bickle's attention to the micro, to "the rest" of neuroscience. We disagree with Bickle's conclusions, though. He argues for a "ruthless reductionism", as he puts it, "from minds to molecules", and cites evidence that neuroscientists themselves also see things this way. We agree that neuroscientists tend to be ruthless reductionists, but we have become convinced that this is not necessarily a good thing. We will show that a close look

at some methodological facts (and foibles) of work in the neurosciences pushes in the opposite direction.

Neuroscientific explanations at the level of molecules and genes will fail if they lose track of the fact they are explanations of molecules and genes in the brains of behaving animals that are situated in and interacting with complex environments. Reductionism, that is, is problematic in both neuroscience and neurophilosophy. Or so we will argue. We will argue for this via a critical review of a portion of the neuroscientific literature and then by presenting some data from one of our own experiments.

Before we begin in earnest, though, it is worth pausing a moment to make our intentions clear, particularly concerning the intentions we do not have. Although we disagree with Bickle's reductionism, we agree with his "metascience" approach (more on which below), and do not wish to argue by invoking normative or *a priori* principles. We are not proposing or endorsing dualism, non-reductive physicalism, or multiple realizability. We will instead be arguing that the proper object of study of a scientific psychology is spatio-temporally larger than the brain. This makes us anti-reductionists, of course, but we are anti-reductionists who became such by virtue of empirical evidence and careful attention to methodology. (See Gibson 1979; Varela et al. 1991; Clark 1997, 2003; Varela and Thompson 2001 for unrelated, though similarly empirical, arguments against reductionsim.) We do not wish to be confused with those who worry about possible worlds, varieties of supervenience, Martian psychology, or zombies.

2. REDUCTION, RUTHLESS AND OTHERWISE

We take it that readers of this journal do not need a detailed discussion of reductionism. We will, therefore, describe reductionism only briefly, focusing specifically on reductionism as it related to the mind-body problem. The first point to make is that, following Nagel (1961) and most recent work on reduction, we will be interested in intertheoretic reduction, as opposed to ontological reduction. (See Silberstein 2002 for a discussion of both styles of reductionism.) That is, as we will use the word, reduction is a matter of accounting for the facts, generalizations and laws of one theory, the reduced theory, in terms of the facts, generalizations and laws of a second, more general theory, the base theory, along with some bridge

principles. In a successful reduction, the base theory and the bridge principles entail the reduced theory. Nagel's basic formula has been modified many times over the years (for example by Hooker 1981a, b, c; Churchland 1985, 1989), but it will suffice for present purposes. A key feature of reductionism so conceived is that the base theory must be a more general theory of the same slices of space and time. That is, you have to be able to use the base theory to account for activity in the same regions of space the reduced theory accounts for, and others to boot. Physics, then, is a good candidate for a base theory because its laws and generalizations are thought to apply always and everywhere.

In the arena of the mind-body problem, the ultimate goal of reductionists may be to use physics as a base theory, but only via intermediaries. In today's discussions the proposed reduction is from cognitive science to neuroscience. The guiding metaphor of cognitive science actually encourages the reduction of cognitive science to neuroscience. Cognitive science textbooks (Stillings, et al. 1995; Thagard 1996) present cognitive science as guided by an analogy between the mind and computer programs. The basic idea is that the mind is to the brain as the program is to the computer that is running it. So despite a good deal of anti-reductionist rhetoric on the part of cognitive scientists and philosophers thereof (e.g. Fodor 1974), it is open to the would-be reductionist to accept the analogy and argue as follows: any explanation of the operation of a computer program would be entailed by an analysis of the computer itself, along with some bridge principles (among which would be a story about how the program is compiled, and so on). So too for the science of the mind and the science of the brain. Ultimately, reductionists will argue, the neurosciences will account for the generalizations, laws and facts of cognitive sciences. And indeed, many proponents of the cognitive sciences have made these arguments. The most common strategy employed here is to argue that cognitive science is reducible to *cognitive* neuroscience. This is a natural way to proceed: cognitive scientists typically view mental processes as computational manipulations of representations; cognitive neuroscientists view neural processes as computational manipulations of representations. Paul and Patricia Churchland have been the most famous proponents of such a reduction (P. S. Churchland 1986, 2002; P. M. Churchland 1979, 1989),¹ but see also Bickle 1998 and Bechtel and Mundale 1999. We take it that this style of reductionism is familiar.

More recently, as mentioned in section 1, John Bickle (2003) has argued in favor of a different sort of reductionism, which he calls ruthless reductionism, in which cognitive science is reduced to *molecular*, rather than cognitive, neuroscience. To avoid a raft of problems that would attend such a reduction, including nit-picky issues over the exact nature of reductionism and questions of the content of cognitive states (what happens to it?), Bickle advocates what he calls “new wave metascience”. In new wave metascience, one trusts neuroscientists, who understand neuroscience much better than philosophers after all, to know what their work has accounted for. So rather than focus on the *n*th-level objections and replies to philosophical accounts of intertheoretic reduction, a new wave metascientist focuses on the claims made by scientists themselves, and lets the neuroscientific results speak (in great detail and with impressive felicity, we might add). Such views might seem strange coming from a philosopher, particularly from a philosopher who has contributed to the debate over the exact character of intertheoretic reduction, so it is worth pausing briefly to look at ruthless reductionism in action. Bickle’s demolition of multiple realizability arguments provides a good window.

The basic idea behind multiple realizability arguments is that because cognitive things could be made of material other than brain-stuff, cognitive things are not identical to neural things. (Putnam 1975 includes several of his arguments for multiple realizability.) That angels and ghosts without material brains, robots with metal and silicone brains, and Martians with brains much different from ours could in principle feel pain shows that pain is not identical to a type of brain state. So no reduction, ontological or intertheoretic, of the cognitive realm to the neural is possible. Our (highly fallible) sense is that most philosophers believe that the cognitive is multiply realizable, and hence not reducible to the neural. Bickle’s ruthless reductionist response to multiple realizability comes in two parts. First, the neuroscientist and the new wave metascientist don’t care about multiple realizability, about whether angels or aliens might have differently implemented minds. The issue that matters to the (new wave meta-) scientist is whether real minds on Earth are multiply realized. The issue is not imaginable multiple realizability, but actual multiple realization. Of course, the data concerning whether mental states are multiply realized on Earth are not all in, but some are. For example, the current sense among the neuroscientific community is that long-term memory in mammals just

is long-term potentiation. Bickle uses this consensus to argue that our cognitive scientific theory of long-term memory (especially the consolidation of short term memory and the storage of memories) is reducible to the neuroscientific theory of long-term potentiation. He then shows, in great detail, that the very same molecules (including cAMP, CREB, PKA, for those interested) and mechanisms at work in mammals during long-term potentiation are also at work in long-term potentiation in fruit flies and sea slugs. Bickle takes this truly surprising identity of molecular mechanisms in these evolutionarily distant species to indicate that long-term potentiation and memory consolidation are realized in exactly one way on planet Earth. So LTM, at least, is not multiply realized. Whether this holds of the cognitive in general is an empirical matter, one that has nothing to do with thought experiments. The ruthless reductionist bets that other things studied by the cognitive sciences will be accounted for as LTM has been.

As might be obvious by now, we prefer Bickle's ruthless reductionism for a variety of reasons. First, we mistrust the representational and computational assumptions of both cognitive science and cognitive neuroscience (Chemero 2000, 2001). Second, we cautiously approve of Bickle's "new wave metascience" approach. We like the cleanliness with which it cuts off many kinds of debate, particularly *a priori* philosophical arguments that such-and-such a theory can never account for so-and-so a subject matter. We find these arguments even more deplorable than idle speculations about Martian physiology. Third, and perhaps most importantly, Bickle's book shines a philosophical light on the *rest* of neuroscience, the non-cognitive majority that focuses on cellular, molecular, and genetic mechanisms. Fourth, we should point out that—new wave metascience aside—Bickle's reduction of the cognitive to the molecular is fully in line with the goals of other reductionists; it just goes a step further, casting cognitive science in terms of genetics and biochemistry, which is more general than cognitive neuroscience.

So: Three cheers for ruthless reductionism, which we think is the best hope for reductionism of the cognitive to the neural. We also think Bickle is correct that most neuroscientists are ruthless reductionists. The next several sections are an argument that this ruthless reductionism is actually bad for neuroscience. We will argue for this via a review of research in molecular neuroscience, psychopharmacology and behavioral genetics that uses object exploration by rats and mice to assess the role of particular brain areas,

neurotransmitters, genes, and drugs in learning, memory and intelligence. And then with an object exploration experiment of our own. The point will be to show that object exploration experiments are potentially confounded precisely because those who perform them are ruthless reductionists. This is something that should bother ruthless reductionist philosophers. We will also make recommendations concerning how one might correct the confounds we find in object exploration. These recommendations will amount to an argument against regulation, non-ruthless reductionism.

There will not be much in the way of philosophy in the next several sections. In section 3, we briefly describe the increasingly popular object exploration paradigm in behavioral neuroscience. Section 4 reports the results of a review of the object experiment literature. We find that many of those who use the object exploration paradigm are somewhat cavalier about the types of objects they use. In particular, they pay little attention to the affordances (Gibson 1979) of the objects for the species of animal they are studying. In section 5, we describe one of a series of experiments that shows that mice explore objects differently, depending on the affordances of those objects. In section 6, we present some conclusions, arguing in particular that neuroscientists studying object exploration must let go of the assumption that studying molecules suffices for studying cognition and that they should embrace a more ecological approach. This methodological conclusion has philosophical consequences for ruthless reductionism and for psychoneural reductionism in general.

3. WHY STUDY OBJECT EXPLORATION?

One of the most commonly used behavioral methodologies in animal research is the study of exploratory behavior (Renner 1990; Hughes 1997). Unfortunately, in current practice, exploratory behavior is typically assessed by observing animals in an empty open-field and recording only the number of line crosses and/or rears. We believe that, although useful, this represents a restrictive view of "exploratory behavior". Therefore we have begun to focus on tasks that provide the animals with additional opportunities for exploration (e.g., objects; Chen et al. 2000; Renner and Rosenzweig et al. 1986; Roullet et al. 1997). There are at least two advantages to the object exploration task. First, there is no explicit need for

any food or water restriction. Second, the task is useful for the study of various forms of learning (e.g., habituation and recognition memory).

The general procedure for this task involves an open field apparatus and several objects (two or more). In the case of multiple objects, either identical objects are used or sets of different objects are used within each trial. In either case, the set of objects is repeatedly presented and the behavior of the animal is recorded. After a number of presentations, either one object is replaced with a novel object or the spatial configuration of the objects is changed. In this sense, one can infer recognition memory from the animal's behavioral response to the environmental change. This inference is based on two aspects of behavior. First, rats and mice tend to seek out and explore novelty (with the caveat that the novel feature is not too stressful or aversive). Second, with repeated presentation, the animals become "familiar" with a certain set of objects (i.e., the animals habituate to the objects in terms of their novelty). Therefore, changes in the environment should result in preferential exploration of the newly substituted object or spatially rearranged objects. Experimenters assume that in order to respond to these changes, the rat must remember (i.e., recognize) either the familiar objects or the novel spatial change.

Our initial use of the object exploration task was to assess recognition memory in rats (Renner and Rosenzweig 1986; Roullet et al. 1997; Save et al. 1992). More specifically, we were interested in examining the effects of ethanol withdrawal on recognition memory in rats (Heyser and Chemero, under review). Surprisingly, these results showed that ethanol withdrawal did not impair recognition memory. In fact, ethanol-withdrawn rats reacted with an increase in exploration of novel objects when compared to controls (non-withdrawing rats). Initially, we concluded that perhaps ethanol withdrawal affects the habituation process in these animals. This would have been a striking finding. Imagine being continuously drunk for four weeks, then finding an improved recognition memory during the first few hours of withdrawal. Unfortunately, after closer examination of the data, it became clear that the findings were greatly influenced by the properties of the objects, with rats preferring objects they could climb on to those they could not. That is, the rats showed a preference for objects that have affordances for common rat activities.

4. RECENT WORK ON OBJECT EXPLORATION

The results of our initial study led us to investigate whether other studies had reported similar results. In reviewing a portion of the extensive literature that has been published using object exploration/recognition, we found that very little detail is provided concerning the nature of the objects selected for these tasks. Perhaps most disturbing is the fact that there is little to no discussion of why or how the particular objects were selected for use in these experiments. This lack of detail concerning the objects used in studies is unsurprising, for two reasons. First, consider the discussion of the objects used in the first recent object exploration paper, Ennaceur and Delacour 1988. This paper, which has been cited more than 150 times, has only the following as a description of the objects used: "The objects to be discriminated were made of glass, plastic or metal and existed in duplicate. Their weight was such that they could not be displaced by rats." (48) This description of the objects is repeated verbatim (or nearly verbatim) in all of the object exploration studies done by Ennaceur (and various colleagues), one or more of which is cited by every recent object exploration study we have encountered. Second, researchers who use the object exploration task are typically molecular neuroscientists, behavioral geneticists, and psychopharmacologists; that is, they are exactly the types of neuroscientists who, as Bickle points out, tend to be ruthless reductionists. Yet in spite of this lack of concern for to-be-explored objects, the nature of the objects used in these studies may be critical to the behavior observed (i.e., initial exploration, response to novelty, and recognition memory) in that different objects afford different activities. For example, some objects can only be touched or reared on, whereas other objects can be climbed onto and into, leading to potentially *qualitatively* and *quantitatively* different exploratory behaviors (Renner and Rosenzweig 1986).

We reviewed 116 recent papers on object exploration published in neuroscience journals.² For each article, we attempted to determine the age(s) and species of animal used and what the properties of the object used were. We used this information to determine what the *affordances* of the to-be-explored objects were for the animals in the studies. Affordances, introduced by Gibson (1979) are opportunities for action that exist in the environment and are perceived directly (i.e., non-inferentially) by animals. There has been significant disagreement over the precise nature of affordances: some have

argued that affordances are categorical properties of environmental objects (Reed 1996); others have argued that affordances are dispositional properties of environmental objects (Turvey 1992, Scarantino 2003); still others have argued that affordances are relations between animals and environments (Sanders 1997; Chemero 2003; Stoffregen 2003). It is beyond the scope of this essay to settle this matter, but for our purposes it is easiest to assume the third possibility. If we assume that affordances are relations between the abilities of animals and the properties of objects, then two pieces of information (species and object property) allow us to determine the affordances of the objects for the species used in the study. Crucially for our literature review, we could also determine whether different objects used in individual studies have equivalent affordances.

We stipulate that objects have equivalent affordances whenever the relationship between some animal's abilities and the objects are sufficiently similar that the animal can perform the same stereotypical behavior with respect to the object. We also assume that "same behavior" is read very broadly, so that climbing (pushing, hiding-in, etc.) of any kind is the same behavior. To take one relevant, illustrative example, any two objects that

- (1) are sufficiently sturdy to support the full weight of the animal without deformation, and
- (2) have a surface that is
 - a. parallel to the ground,
 - b. at a height less than the full length of the animal (tail excluded),
 - c. at least as wide as the distance between the animal's left and right feet, and
 - d. at least as long as the distance between the animal's front and back feet

afford the same behavior: climbing-on. This rather loose individuation of behaviors, and hence of affordances, allows us to say that a copy of *Being and Time* and a Palm Pilot, but not a beer bottle, afford climbing-on for mice and rats. It does not make a distinction between the different sequences of muscle movements required to get the animal onto these objects.

The main results of the literature survey were as follows. Of these 116 articles, 52 (approximately 44%) gave little or no information concerning the specific objects that were given for exploration.

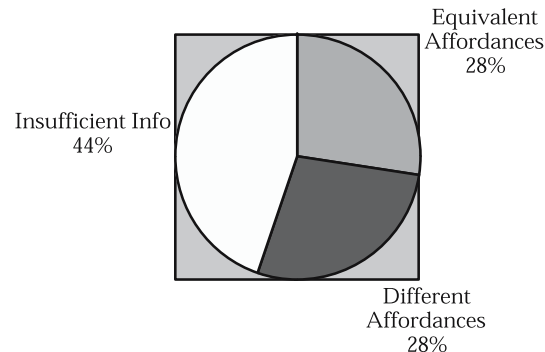


Figure 1.

64 articles (approximately 56%) provided detailed descriptions of the objects. Of the 64 articles that included descriptions of the objects used in the experiments, 32 (approximately 28% of the total) used sets of objects with non-equivalent affordances for the species of animal under study. (For example, several studies compared exploration of objects that could be climbed by rats with objects that could not be climbed.) Only the remaining 32 studies used objects with similar affordances for the animals under study. (See Figure 1.)

This leads to the following evaluations of the current state of research on object exploration. Approximately 72% of the research papers we reviewed may have the same methodological problem we found in our initial work, making the results difficult to interpret and making it virtually impossible to generalize across experiments. Call these the *potentially confounded studies*. The majority of these potentially confounded studies (44% of the articles we surveyed) do not give enough information about the objects used, and so fail to meet one of the primary goals of scientific research – these studies are *not replicable*. The remainder of the potentially confounded studies is *suspect*, because their results could be skewed by the comparison of exploration of objects that have different affordances. In these studies, there seems to be the attitude that it doesn't really matter what the objects are, as long as one or more of them are novel. The objects were either deemed not worth describing or were simply things that happened to be around the lab (e.g., soda cans are popular).

We can see, at least provisionally, that the objects used do in fact matter by looking at one particular study: Morrow et al. "Prenatal cocaine exposure disrupts non-spatial, short-term memory in

adolescent and adult male rats” (2002). We pick on this study for two reasons. First, because it has apparent implications for human parenting, it received mainstream press attention. Second, this study presents its data on object exploration in a unique and commendable way. Before getting to this uniqueness, we should note that the Morrow et al. write up says nothing specific about the objects used. Here is the one sentence that mentions, in passing, the objects used: “On test days . . . the rats were placed into an open field with two identical objects for 3 min and, after a delay (20 min, 1 or 24 h) were returned to a duplicate open field and allowed to explore two objects, one identical to the original objects and the other different.” (p. 218) It is evident that Morrow et al. did not think the nature of the objects to be important. Yet their data presentation gives separate exploration times for each of the objects used in the study. The data, which are quite telling, are presented in Table I. There are two key features of this table for our purposes. The first is object C. The mean exploration time for object C is much lower than that of any other object. Second, object D differs from all the other objects in terms of variance as reflected by the standard error of the mean, especially in the saline condition (mean exploration time: 12.90 ± 7.30 seconds). These objects, and especially object C, were clearly explored differently than the other objects used in the study. But without any further description of the objects, it is impossible to tell why this occurred. Furthermore, although they presented this data on the exploration of individual objects, Morrow et al. clearly didn’t think them important. For the rest of the statistics in the paper, exploration times are averaged across objects. There is reason to think that this may impact the results and specifically the conclusions drawn from this study. Morrow et al. surely got some significant effects, it is just not clear whether those effects are due to prenatal cocaine exposure (as their title indicates) or to additional variance introduced by differences in the to-be-explored objects, or some combination of these. The same worry attends 72% of the object exploration studies we reviewed.

This rather pessimistic evaluation has led us to do a series of experiments on object exploration and affordances. These experiments are designed to explore the relationship between the affordances of objects and the ways that animals (mice in our case) explore them. The pessimism of our evaluation is unfounded if our experiments show that animals explore objects similarly, whether or not they have different affordances. The experiment also has

TABLE I

Part of Table 2 in Morrow et al. (2002)

Object	Saline Condition	Cocaine Condition
Mean exploration times, in seconds		
A	14.93 $\pm$ 4.25	19.20 $\pm$ 3.49
B	11.01 $\pm$ 2.77	12.75 $\pm$ 3.19
C	6.89 $\pm$ 1.97	7.48 $\pm$ 2.77
D	12.90 $\pm$ 7.30	10.97 $\pm$ 2.69

consequences for ruthless reductionism. If scientific experiments suffer because ruthless reductionist neuroscientists attend to neurochemicals to the exclusion of other behavior-relevant issues (i.e., the affordances of the objects they use), it would seem that ruthless reductionism among neuroscientists is counter-productive. This would be a problem for ruthless reductionism.

5. ARE ALL OBJECTS EQUAL? AN EXPERIMENT

We have recently begun a series of studies with mice specifically looking at object selection and its impact on performance in the object exploration task. (We describe the experiment reported here, and several others in more detail in Heyser and Chemero, under review.) For this study, we used adult male and female C57BL/6 mice, randomly assigned to two groups. The first group (TOUCH) was tested with objects that could not be climbed. That is, the objects did not meet the criteria for affording the behavior climbing-on described above. The second group (CLIMB) was tested with objects that did meet these criteria. (See Figure 2.) Testing was conducted in a large circular arena (large, that is, for a mouse – 1.2m in diameter) and consisted of four 6-minute trials, with a 3-minute intertrial interval between each trial. There were no objects in the arena during Trial 1, which served as a baseline/familiarization period to the novel environment. The number of line crosses and rears were recorded. In Trial 2, four different objects were placed in the open field. For mice in the TOUCH group, these objects consisted of a glass bottle, a metal rectangular can, a set of plastic stacking squares, and a plastic toy barrel. All objects were of sufficient height and weight so that they could not be climbed or moved. In addition, the objects did not afford chewing and were

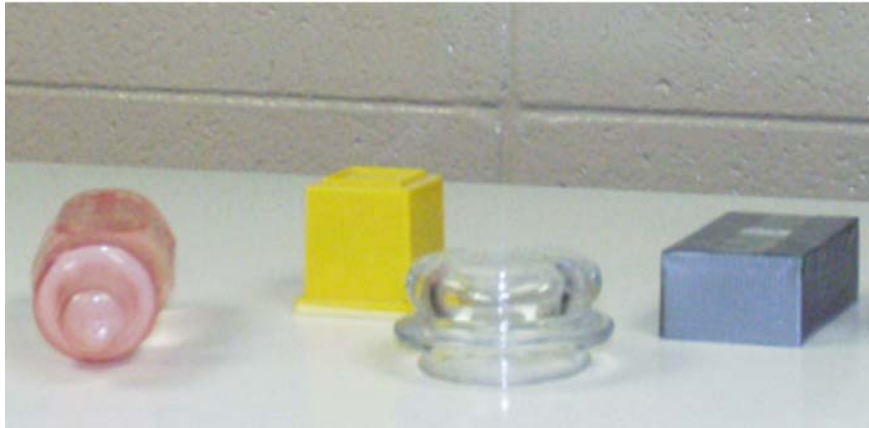
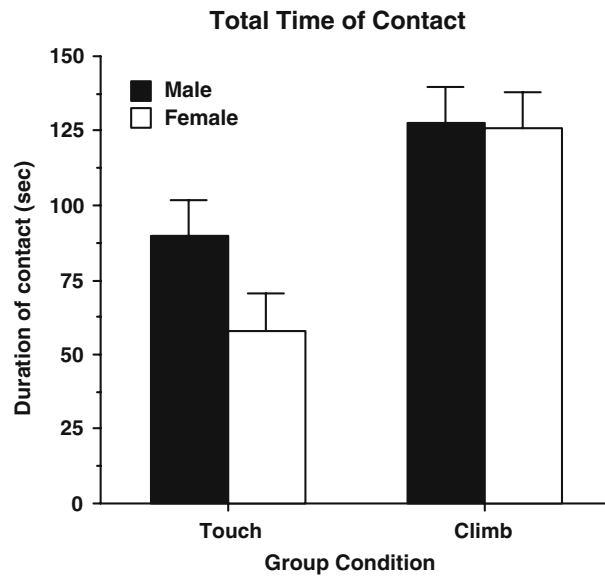
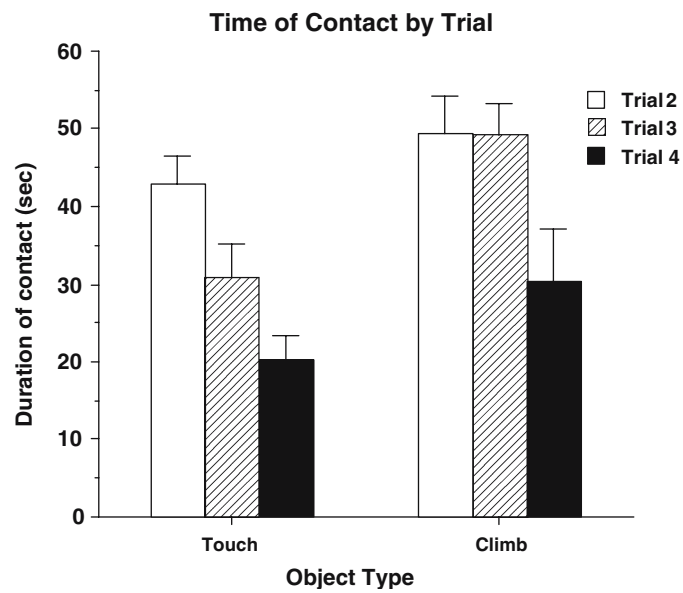


Figure 2. Objects that afford climbing for C57/B16 mice.

washed between trials. For mice in the CLIMB group, the objects were a plastic square, a rectangular bottle on its side, a cardboard box, and a plastic object with a hole in the center. All objects were of sufficient height and weight so that they could be climbed on, but not moved. As above, we washed the objects between trials, and they did not afford chewing. The objects were positioned equidistant from the walls and from each other. The same four objects remained in the open field during Trials 3 and 4. Several behaviors were recorded during Trials 2 – 4 including: line crosses, rears, grooming, the frequency and duration of contact with each object, and the latency to make first contact with each object within a given trial.

We found clear differences in exploratory behavior as a function of the type of object. First, mice in the CLIMB group spent significantly more time in contact with the objects, when compared with duration scores in the TOUCH only group (see Figure 3). Thus, objects that could be climbed on were explored longer. The type of object also influenced the rate of habituation. Specifically, the mice in the TOUCH group showed faster habituation (as defined by a decrease in the duration of exploration over time) than mice in the CLIMB group (see Figure 4).

Put simply, this experiment shows that all objects are not created equal: their affordances really do affect the way animals explore them. This means that the assumption – common among molecular neuroscientists, behavioral geneticists and psychopharmacologists –

*Figure 3.**Figure 4.*

that studying neurochemicals can be a replacement for studying behavior is harmful, at least when one uses the object exploration methodology.

6. CONCLUSIONS

There are several lessons to learn from the preceding, some neuroscientific and some philosophical.

First, our experiment shows that object exploration studies cannot use just any objects. It also shows that our earlier pessimistic evaluation of the current literature on object exploration does in fact hold up. The differences in the exploratory behavior that mice display given the affordances of the objects to be explored indicate that at least 32, and as many as 84, of the 116 articles we survey report results that should be seen as potentially confounded. There are, of course, more than 116 published articles on object exploration, but we bet that these numbers will hold up for all of the papers on object exploration. If we're right, nearly 3/4 of the literature on object exploration is potentially confounded. It is not our contention that the data obtained in these studies are not "real". However, given that these studies rely on a simultaneous presentation of multiple objects, it is possible (even highly likely) that the behavior of the animal is not only influenced by the affordances of a specific object but also by the interaction of all available objects. This point is most problematic in situations where the objects are explored differently (i.e., when the affordances of the objects are not equivalent). Therefore, although the data from these experiments are not flawed *per se*, the conclusions drawn from these experiments may be suspect and at the very least give rise to many alternative hypotheses.

Second, journal reviewers and editors must insist that methods sections describe experimental procedures sufficiently that the experiments can be replicated. This may seem obvious, but recent trends are for abbreviated methods sections. For example, the strict word limits of high impact journals like *Science* and *Nature* increase pressure for skeletal descriptions of experimental methods. Given the quite deserved respect that these journals have, the tendency to skimp on description of methods is spreading to other journals (and not just in the neurosciences). In the worst cases of this, methods sections are primarily citations of methods sections in other papers.

This practice simply must end. We hope that the increasing reliance on electronic media in academic publication will loosen page and word limits, and in so doing will reverse this trend.

Third, we hope that the experiment presented here, along with several others in progress, will be taken as providing the necessary background information required to do object exploration experiments more effectively. For example, if one wants to use differences in object exploration to indicate that some gene or neurotransmitter is crucially involved in memory, it is vital to use objects that, all other things being equal, animals explore in the same way. The experiment here is the beginning of a catalog of the exploration preferences of mice.

Fourth, the problem that we have diagnosed with the extant research on object exploration is methodological, but we think a conceptual issue lurks not far behind the scenes. The problem is that neuroscientists tend to ignore the animals attached to the brains they are interested in studying. The solution to this problem is to adopt an understanding of animals that insists on studying brains, perception, memory, cognition, etc. only in terms of their relation to behavior. We recommend the ecological approach as described by Gibson (1966, 1979) and his followers. According to this approach to psychology, one cannot understand perception or cognition separately from action. Our literature review and experiment described above are an example of the benefits that might come with adopting such an approach. We have shown that failing to attend to the species-typical behaviors of mice and rats and whether or not the objects to be explored afford any of those behaviors has led to a series of results that could be artifacts of the experimental setup. A mouse or rat may recognize that some object is novel, indicating that it remembers what objects were present before, but may not explore that object because lacks affordances for species-specific behaviors. Although, we have not done the same sort of systematic literature review or experiment on any other research methodology, we would be unsurprised to find similar artifacts in other experimental paradigms, such as gaze preference in human infants and non-human primates, that make generalizations about learning (or cognition or memory) based upon the amount of time an animal spends attending to something. In these fields as well, the ecological approach – particularly, careful attention to affordances – could turn out to be essential.

Getting now to the more philosophical conclusions, we think our results indicate that ruthless reductionism can lead to bad neuroscience. The work on object exploration that we surveyed may have suffered because experimenters were not as concerned about the nature of objects they used, despite the fact that those objects can have profound effect on exploratory behavior. Clearly, the molecular neuroscientists, behavioral geneticists and psychopharmacologists thought that the objects did not matter. We attribute this neglect to the attitude among these researchers that studying molecules *is* studying cognition; that is, we attribute the choice of objects in these experiments to the experimenters' ruthless reductionism. This, we take it, is a serious problem for the ruthless reductionist. A philosophical attitude toward science that leads to potentially confounded science is not an attitude that scientists ought to hold. Remember that ruthless reductionism is connected with new wave metascience, in which philosophical attitudes toward scientific research take their cues from the attitudes of the scientists whose work it is. Our literature review and experiment indicate that neuroscientists who use object exploration may need to reflect on and empirically validate the process of object selection, and realize that behavior and the affordances that specify it matter. So ruthless reductionism is an inappropriate position for neuroscientists to hold because ruthless reductionism can lead to questionable experimental methodology. Ruthless reductionism, then, is also inappropriate for philosophers of science.

Sixth, the methodological changes required to improve object exploration experiments have more general consequences for reductionism about the cognitive. As noted above, much cognitive psychology and cognitive science encourages reductionism by comparing the mind to software running on the brain. Spatially, then, the object of study of cognitive psychology and neuroscience are the same; these two theories just focus on different "levels". But our recommendation for fixing the methodological problems with object exploration research involves adopting an approach to cognition in which the object of study is spatially larger than the brain. According to the ecological approach, the object of study for cognitive psychology is the whole animal plus aspects of its environment, what is often called the "animal–environment system". To the extent that ecological psychologists study brains, they study them only as parts of behaving animals in information-rich environments. This makes a reduction of cognitive psychology to neuroscience (cognitive or

molecular) out of the question. Researchers in lower-level theories such as neuroscience depend on higher-level theories (e.g., from cognitive psychology) to help them decide what needs explaining. For example, we can only reduce LTM to LTP if we have a reasonable theory of LTM to guide research on brains and to be reduced to a theory of LTP. Neuroscientists and reductionists, in other words, need psychologists to get things (roughly) right. Our experiment and literature review indicate that getting things even roughly right in psychology (and neuroscience for that matter) requires knowing about affordances. And this is the rub. For in an intertheoretic reduction, the base theory must be a more general theory that applies to the same regions of space and time as the reduced theory. But no theory of the brain alone will be applicable to the brain-body-environment system studied by ecological psychologists. The base theory in this case is not a more general theory of the same subject matter for there are no genes or neurotransmitters in the majority of the system under study. So reduction of cognitive psychology to neuroscience will fail.

7. FINAL WORDS

We should point out that the reductionist might sensibly point out that we have only shown a problem with one sort of experiment, and that reductionism has served neuroscientists very well in other settings, e.g., in the work on memory that Bickle cites. But to turn this into a defense of reductionism, it must be argued that the work we review on object exploration is an exception to otherwise methodologically unimpeachable work done by other ruthless reductionist neuroscientists. Then one could argue that, in general, neuroscientific work is not aided by expanding the focus of inquiry to include non-neural (so non-neurochemical) things such as bodies and affordances. The only argument we can imagine for this would be careful study and defense of other methods used by molecular neuroscientists. We think that both neuroscience and neurophilosophy would benefit if philosophers were to pay this sort of close attention to additional experimental methodologies. Indeed, we would be happy to have this work done, even should it be the means by which reductionism is vindicated and we are proven wrong.

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NOTES

¹ The Churchlands are even more famous for being eliminativists about folk psychology. Their guiding idea is that the cognitive science that is ultimately reduced to cognitive neuroscience will make folk psychology a predictive and explanatory embarrassment. They are *not* eliminativists about cognitive science.

² We have not included the citations for these papers. The list of full citations for the 116 articles reviewed is longer than the present essay.

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